

Cloud Native Streaming Platform Netflix-inspired

Santiago Antonio Gualteros Bustos

Nicolas Castro Rivera

Andrés Julian Vargas Medina

Erick Santiago Buitrago Peña

1 Executive Summary

Netflix is a streaming platform that provides this service for subscriptions. Whose vision is entertaining customers through series and movies. This content can be accessed around the world by millions of customers those who want to watch their favorites shows at any time. For that reason is an application demands great availability, good performance with heavy content such as multimedia and personalized experience with assertive recommendations.

2 System Description

Business problem, actor catalog, use case definitions, and system boundary diagram

2.1 Business Problem

Netflix solves the problem of providing a large amount of audiovisual content to millions of users through the internet while giving each user a convenient and personalized experience.

Users want to watch movies and series whenever and wherever they want without having to download the entire content beforehand. At the same time, Netflix needs to help users find something they want to watch among a very large catalog.

The platform provide another services that not will be consider in this project like: Option for subtitles, interactive experiences or content preview.

2.2 Actors

The application has 4 users represented by a person, each of this person has a particular function and use the application for different purposes. Also there are some system actor that provide services for the application.

Table 1: Actors, Roles, Interaction Frequency, and Expectations

Actor	Role	Frequency of interaction	Expectation
Customer	A person who has a monthly membership plan with the platform.	High	Be able to watch series and movies without interruptions and receive relevant recommendations.
Operational user	A member of the operations team responsible for managing content within the system.	Medium	Be able to manage all the content without problems.
Data analyst	A member of the Data & Insights area who can access data and statistics.	Medium	Have access to useful data with clear statistics.
Platform administrator	Is responsible for managing users and the business model.	Low	Be able to manage the users of the system and membership configurations.
Media Convert service	It allows videos to be sent in small segments.	High	The user does not need to download the entire video; it is stored in fragments in the device buffer.
CDN service	Servers that help reduce latency by delivering content from the nearest server.	High	Ensure rapid response to high volumes of requests.
Storage	Storage for multimedia content and metadata, useful for performing queries.	High	Persist all data over time.

2.3 Main Use Cases

Subsection example 2

““latex

Table 2: Use Case CU 1.1 – Sign up in app as a customer

Use case	CU 1.1 – Sign up in app as a customer
Actors	Customer
Requirements	RF-01, RF-02
Precondition	The user does not have a registered account in the system and wishes to become a customer.
Main flow	
Actor	1. Accesses the registration screen and selects the "Sign up" option.
System	2. Displays the registration form requesting a Gmail account (or email) and password, along with the available membership plans.
Actor	3. Selects a membership plan and enters the requested account credentials.
System	4. Requests the payment method information (card or payment gateway) to process the membership.
Actor	5. Enters the payment details and confirms the payment.
System	6. Sends the payment information to the payment gateway for processing.
System	7. Validates the payment confirmation and creates the customer account.
System	8. Sends a notification confirming the account creation and active membership.
Alternative flow – Payment not accepted	
System	1. Detects that the payment gateway rejected the transaction.
System	2. Displays an error message indicating that the payment was not accepted.
Actor	3. Corrects the payment information or selects another payment method.
System	4. If no valid payment is provided, cancels the registration process and does not create the account.
Postcondition	The customer account is created with an active membership, or no account is created if the payment fails.

Table 3: Use Case CU 1.2 – Log in app

Use case	CU 1.2 – Log in app
Actors	Customer, Operational user, Data analyst, Platform administrator
Requirements	RF-03
Precondition	The user has a previously registered account with valid credentials.
Main flow	
Actor	1. Accesses the login screen.
System	2. Displays the fields to enter the Gmail account (or email) and password.
Actor	3. Enters the credentials and presses the "Log in" button.
System	4. Validates the entered credentials against the stored records.
System	5. Grants access and redirects the user to the dashboard corresponding to their role.
Alternative flow – Incorrect credentials	
System	1. Detects that the email or password entered is incorrect.
System	2. Displays an error message and increases the failed attempt counter.
Actor	3. Re-enters the credentials to try again.
System	4. If the number of failed attempts reaches 5, blocks access to the account temporarily.
Postcondition	The user is authenticated and redirected to their corresponding dashboard, or the account is temporarily blocked after 5 failed attempts.

Table 4: Use Case CU 1.3 – Update user information

Use case	CU 1.3 – Update user information
Actors	Customer, Operational user, Data analyst, Platform administrator
Requirements	RF-06, RF-04
Precondition	The user is authenticated in the system with an active session.
Main flow	
Actor	1. Accesses the "My profile" or "Account settings" section.
System	2. Displays the user's current personal information and access credentials.
Actor	3. Selects the field(s) to edit (email, password, or personal information) and enters the new values.
Actor	4. Presses the "Save changes" button.
System	5. Validates the format and consistency of the entered data.
System	6. Updates the information in the database and displays a success confirmation message.
Alternative flow – Invalid input	
System	1. Detects that one or more fields do not meet the required validation (invalid format, empty field, mismatched password, etc.).
System	2. Displays an error message specifying the invalid field(s).
Actor	3. Corrects the indicated data and resubmits the request.
Postcondition	The user's profile and/or access information is successfully updated, or remains unchanged if the validation fails.

Table 5: Use Case CU 1.4 – Cancel membership

Use case	CU 1.4 – Cancel membership
Actors	Customer
Requirements	RF-05
Precondition	The customer has an active membership and is authenticated in the system.
Main flow	
Actor	1. Accesses the "Manage membership" section and selects "Cancel membership".
System	2. Requests confirmation of the cancellation and the reason for it.
Actor	3. Confirms the cancellation and provides the requested reason.
System	4. Processes the cancellation request.
System	5. Updates the membership status to inactive and displays a confirmation notice.
Alternative flow – Login attempt after cancellation	
Actor	1. Attempts to log in after the membership has been cancelled.
System	2. Detects that the membership is inactive and denies access to the platform's content.
System	3. Notifies the customer that a new payment is required to reactivate the membership.
Postcondition	The membership is cancelled and marked as inactive; access to the platform remains restricted until the membership is renewed.

Table 6: Use Case CU 1.5 – Search content

Use case	CU 1.5 – Search content
Actors	Customer
Requirements	RF-08, RF-12
Precondition	The customer is authenticated and has an active membership.
Main flow	
Actor	1. Accesses the main screen and views the personalized recommendation feed.
System	2. Generates and displays content recommendations based on watch history, contextual data, and previous searches.
Actor	3. Enters a search query in the search bar or selects a title from the recommended feed.
System	4. Searches the content catalog for titles matching the query.
System	5. Displays the list of matching movies/series along with their metadata (title, genre, synopsis, image).
Alternative flow – No exact match found	
System	1. Detects that there are no exact matches for the entered search query.
System	2. Generates and displays a list of similar or related content based on the recommendation engine.
Actor	3. Selects one of the suggested titles to continue.
Postcondition	The customer receives a list of content matching or related to their search, ready to be selected for playback.

Table 7: Use Case CU 1.6 – Plays content

Use case	CU 1.6 – Plays content
Actors	Customer
Requirements	RF-07
Precondition	The customer is authenticated, has an active membership, and has selected a movie or episode from the catalog.
Main flow	
Actor	1. Selects the movie or episode to watch and presses "Play".
System	2. Loads the selected content and starts instant video streaming without requiring a prior download.
System	3. Displays the playback controls (play, pause, rewind, forward, volume).
Actor	4. Watches the content, using the playback controls as needed.
Alternative flow 1 – Video streaming failure	
System	1. Detects a failure in the video stream (e.g., corrupted file or server error).
System	2. Pauses the playback and displays an error message explaining the issue.
Actor	3. Chooses to retry playback or select different content.
Alternative flow 2 – Unstable internet connection	
System	1. Detects that the user's internet connection is slow or unstable.
System	2. Displays a buffering/loading indicator while attempting to recover the stream.
Actor	3. Waits for the stream to resume or checks their internet connection.
Postcondition	The content is streamed successfully and watched by the customer, or playback is interrupted and the customer is properly notified of the issue.

Table 8: Use Case CU 1.7 – Manage content

Use case	CU 1.7 – Manage content
Actors	Operational user
Requirements	RF-09
Precondition	The operational user is authenticated and has permissions to manage the content catalog.
Main flow – Upload new content	
Actor	1. Accesses the content management module and selects "Add new content".
System	2. Displays a form to upload the video file and enter its metadata (title, genre, description, cast, duration).
Actor	3. Uploads the video file and completes the requested metadata.
System	4. Validates the uploaded information and stores the new content in the catalog.
System	5. Displays a confirmation message and updates the catalog state.
Alternative flow 1 – Edit existing content	
Actor	1. Selects a specific video already registered in the catalog.
Actor	2. Modifies the video file and/or its metadata.
System	3. Validates and updates the record, then confirms the change and refreshes the catalog state.
Alternative flow 2 – Delete content	
Actor	1. Selects a specific video and requests its deletion.
System	2. Performs a logical deletion, marking the video as inactive without losing its metadata.
System	3. Confirms the deletion and updates the catalog view.
Postcondition	The content catalog is updated with the new, modified, or logically deleted item, without permanent loss of metadata.

Table 9: Use Case CU 1.8 – Manage users

Use case	CU 1.8 – Manage users
Actors	Platform administrator
Requirements	RF-10
Precondition	The platform administrator is authenticated and has permissions to manage user accounts and roles.
Main flow – Register a new internal user	
Actor	1. Accesses the user management module and selects "Add user".
System	2. Displays a form to register the user's data and assign a role (operational user, data analyst, or administrator).
Actor	3. Enters the user's information and selects the corresponding role.
System	4. Creates the account, assigns the selected role, and confirms the operation.
Alternative flow 1 – Change a user's role	
Actor	1. Selects an existing user from the list.
Actor	2. Changes the role currently assigned to the user.
System	3. Updates the user's role and permissions, then confirms the change.
Alternative flow 2 – Ban a user	
Actor	1. Selects a user and requests to ban the account.
System	2. Requests a reason for the ban.
Actor	3. Enters the reason for banning the user.
System	4. Disables the user's account, records the reason, and confirms the action.
Postcondition	The user account is created, its role is updated, or the account is banned according to the administrator's action.

Table 10: Use Case CU 1.9 – View reports

Use case	CU 1.9 – View reports
Actors	Data analyst
Requirements	RF-11
Precondition	The data analyst is authenticated and has access to the analytics module.
Main flow	
Actor	1. Accesses the reports/analytics module.
System	2. Displays the available report types and filter options (time range, metric).
Actor	3. Selects the desired report and configures the filter parameters.
System	4. Processes the streaming usage data according to the selected filters.
System	5. Displays the processed analytical report with the corresponding metrics.
Alternative flow – Report cannot be displayed	
System	1. Detects an error while processing or generating the requested report.
System	2. Notifies the data analyst that the report could not be displayed.
Actor	3. Retries the request or selects a different report/filter configuration.
Postcondition	The requested analytical report is displayed to the data analyst, or the analyst is notified that it could not be generated.

““

2.4 System Boundaries

Subsection example 2

3 Functional Requirements

This requirements specify what application does. There are 12 functional requirements divided in 7 domains. To define the functional scope for this section, an analysis of Netflix’s business model and official domain vision was conducted to establish a comprehensive understanding of the operational context. Building upon this business domain baseline, an actor-driven requirement elicitation process was executed. System actors, ranging from end-customers to operational staff and data analysts, were identified and evaluated based on their operational responsibilities and system interactions. Each actor’s functional needs, such as secure authentication, uninterrupted media streaming, content curation, and platform analytics, were systematically mapped to define the required system capabilities.

Subsequently, Domain-Driven Design (DDD) strategic design principles were applied to analyze and categorize the elicited requirements. The requirements were grouped into distinct Bounded Contexts,

acknowledging that domain entities and ubiquitous language terms carry specialized business semantics and rules depending on their contextual boundary. Through this domain decomposition, the following core Bounded Contexts were identified:

- **Identity and Access Management (IAM):** Manages user accounts, authentication credentials, and access control policies across the platform.
- **Subscription Management:** Governs membership plans, billing processing, and subscription lifecycle states.
- **Streaming Core:** Serves as the primary domain service responsible for video playback orchestration and real-time media streaming delivery.
- **Recommendation Engine:** Leverages user consumption history, search queries, and contextual metadata to generate personalized content suggestions.
- **Content Management:** Facilitates the complete lifecycle management (CRUD operations) for media assets and metadata across staging and release workflows.
- **Data & Insights:** Captures, aggregates, and processes platform-wide performance and streaming metrics for analytical insights.
- **Catalog & Discovery:** Handles content indexing, search queries, and catalog exploration to enable seamless user discovery.

“latex

Table 11: System Requirements Matrix Categorized by Domain / Bounded Context

ID	Requirement (The system shall...)	Inputs	Outputs	Domain / Context
RF - 01	Allow customer registration through paid membership.	Email, password, payment details	Customer account creation notification	Subscription Management
RF - 02	Allow membership payment processing using online payment methods.	Payment method info (card/gateway)	Payment confirmation or error message	(Subscription)
RF - 05	Allow customers to cancel their active membership subscription.	Cancellation request and reason	Membership cancellation status confirmation	(Subscription)

Continued on next page

ID	Requirement (The system shall...)	Inputs	Outputs	Domain / Context
RF - 03	Allow any user to authenticate using email and password credentials.	Email and password credentials	Access granted to appropriate user dashboard	Identity & Access Management
RF - 04	Allow users to update their access credentials.	Registered email, current and new credentials	Credential update success or failure notice	(IAM)
RF - 06	Allow any user to update their personal profile information.	Authenticated session, updated attributes	Updated profile status or error message	(IAM)
RF - 10	Allow platform administrators to manage user accounts and roles.	Admin request parameters and action payload	Success/failure message and updated user state	(IAM)
RF - 07	Allow customers to stream movie or episode content instantly without prior downloading.	Chapter/movie selection	Instant video playback stream	Streaming Core
RF - 08	Provide personalized and timely content recommendations to customers.	Contextual data, watch history, search queries	Personalized content feed on main screen	Recommendation Engine
RF - 09	Allow operations personnel to manage media content catalog items.	Metadata creation, update, query, or deletion requests	Action result message and updated catalog state	Content Management
RF - 11	Allow data analysts to query and view streaming usage analytics.	Time range and metric filter parameters	Processed analytical reports and metrics	Data & Insights
RF - 12	Allow customers to search for movies or series in the catalog.	Search query string	List of matching movies/series and metadata	Catalog & Discovery

4 Non-Functional Requirements

The non-functional requirements (NFRs) defined for the system are based primarily on publicly available information regarding the Netflix platform and its operational characteristics. The purpose of these requirements is to establish a realistic reference workload and performance context for the cloud-native prototype developed in this project. Some of the values presented correspond to reported Netflix figures,

while others are estimates derived from those publicly available values using explicit calculations and assumptions.

In cases where an official value was not directly available, an estimate was derived from the available information rather than introducing an arbitrary value. This approach allows the prototype to preserve a relationship with the real operational scale of Netflix while maintaining transparency regarding which values are directly documented and which are calculated approximations.

It is important to emphasize that the functional prototype developed for this project will not reproduce the full production scale of Netflix.

If the prototype remains stable under the simulated workload and demonstrates the expected behavior in terms of concurrency, request processing, availability, data management, and scalability, the architecture can subsequently be scaled horizontally by increasing the computational and infrastructure resources provided by the cloud platform.

The objective is not to claim that the prototype can directly sustain Netflix-scale traffic with academic-level infrastructure, but to demonstrate that the system can maintain stability under representative workloads and that its architecture provides a practical foundation for increasing capacity through horizontal scaling as additional cloud resources become available.

Table 12: RNF – Expected Users

Identifier	Description	RF associated
RNF-01	The system shall support up to 325,000,000 registered customer accounts while maintaining normal operation and the defined performance targets.	RF-01, RF-03
RNF-02	Assuming the 96 billion hours viewed in 2H 2025 were uniformly distributed across its 184 days: $\frac{96,000,000,000 \text{ h}}{184 \times 24 \text{ h}} \approx 21.74 \text{ million concurrent users}$ Therefore, the system shall support at least 22 million concurrent users under normal conditions.	RF-03, RF-07, RF-08, RF-12
RNF-03	Peak Concurrent Users: 65 million users. Estimate based on the highest observed peak, not an officially stated maximum capacity.	RF-03, RF-07, RF-08, RF-12

Table 13: RNF – Average Requests per Hour

Identifier	Description	RF associated
RNF-04	The system shall support approximately 50 calls per member per day. Applying this ratio to 325 million members gives approximately 16.25 billion calls per day, equivalent to about 677 million calls per hour	RF-03, RF-06, RF-08, RF-09, RF-10, RF-12

Table 14: RNF – Peak Requests per Hour

Identifier	Description	RF associated
RNF-06	The system shall support at least 600.000.000 application requests per hour during peak periods, equivalent to approximately 160.000 requests per second.	RF-03, RF-07, RF-08, RF-12
RNF-07	The system shall maintain successful processing of authentication, catalog, search, recommendation, and content-access requests during peak periods without exceeding the defined response-time thresholds.	RF-03, RF-07, RF-08, RF-12

Table 15: RNF – Peak Duration

Identifier	Description	RF associated
RNF-08	The system shall sustain the defined peak workload continuously for at least 4 hours without exceeding the established response-time or availability thresholds.	RF-03, RF-07, RF-08, RF-12
RNF-09	The system shall support sustained high-demand periods during evenings, weekends, and periods following the publication of new content without requiring manual intervention to maintain the defined service levels.	RF-07, RF-08, RF-09, RF-12

Table 16: RNF – Response-Time Targets

Identifier	Description	RF associated
RNF-10	At least 95% of authentication, profile, subscription, and standard API requests shall receive a response within 500 ms under normal operating conditions.	RF-01, RF-02, RF-03, RF-04, RF-05, RF-06
RNF-11	At least 95% of catalog browsing and content-search requests shall return the initial result set within 1 second under normal operating conditions.	RF-09, RF-12
RNF-12	At least 99% of catalog browsing and content-search requests shall return the initial result set within 2 seconds under normal operating conditions.	RF-09, RF-12
RNF-13	At least 95% of recommendation requests shall return the personalized content feed within 2 seconds under normal operating conditions.	RF-08
RNF-14	At least 95% of successful video playback requests shall initiate playback within 3 seconds, excluding delays attributable to the user's Internet connection or external content-delivery infrastructure.	RF-07

Table 17: RNF – Availability

Identifier	Description	RF associated
RNF-15	The system shall maintain a monthly availability of at least 99.9% for customer-facing services, excluding previously announced maintenance periods.	RF-03, RF-07, RF-08, RF-09, RF-12
RNF-16	The maximum accumulated unplanned downtime of customer-facing services shall not exceed approximately 43.2 minutes per calendar month.	RF-03, RF-07, RF-08, RF-12
RNF-17	The system shall preserve access to authentication, catalog discovery, and content playback whenever the corresponding service dependencies remain operational.	RF-03, RF-07, RF-12

Table 18: RNF – Data Volume

Identifier	Description	RF associated
RNF-18	The system shall support up to 5 TB of application-managed data at launch, including user accounts, subscriptions, content metadata, viewing history, recommendation data, search records, and analytics data.	RF-01, RF-02, RF-05, RF-06, RF-08, RF-09, RF-11, RF-12
RNF-19	The system shall support a media catalog of at least 5,000 movie and episode entries, including their associated metadata required for catalog discovery and playback.	RF-07, RF-09, RF-12
RNF-20	The system shall allow storage capacity to be expanded beyond the initial 5 TB without requiring a fundamental redesign of the application’s data-management architecture.	RF-07, RF-08, RF-09, RF-11

Table 19: RNF – Data Growth

Identifier	Description	RF associated
RNF-21	The system shall accommodate an estimated annual growth of up to 2 TB of application-managed data without degradation of the defined performance targets.	RF-07, RF-08, RF-09, RF-11, RF-12
RNF-22	The system shall support continuous growth of user activity data, including viewing history, searches, recommendations, and analytics, while maintaining access to data generated during previous years.	RF-08, RF-11, RF-12
RNF-23	The system shall allow storage resources to be increased independently from application services as data volume grows.	RF-09, RF-11

Table 20: RNF – Geographic Scope

Identifier	Description	RF associated
RNF-24	The system shall initially provide its services to users located in Colombia through the public Internet.	RF-01, RF-02, RF-03, RF-07, RF-08, RF-09, RF-12
RNF-25	At least 95% of customer-facing API requests originating from Colombia shall achieve a response time below 500 ms under normal operating conditions, excluding delays caused by the user’s local network or Internet service provider.	RF-03, RF-06, RF-08, RF-09, RF-12
RNF-26	The system architecture shall permit expansion to additional Latin American countries without requiring a fundamental re-design of the core application architecture.	RF-01, RF-02, RF-03, RF-07, RF-08, RF-09, RF-12

5 Workload Model

Structured table with all workload parameters and justification for each value

References

- [1] Rummery, G. A., & Niranjana, M. (1994). On-line Q-learning using connectionist systems. *Technical Report*, Cambridge University Engineering Department.